Kaiqi Yang

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428 S Shaw Ln, Engineering Building RM 3308, East Lansing, MI 48824, USA

Research Interests

- Applications of Large Language Models; Conversational AI; Computational Social Science

Education

- **Michigan State University** *2022 – 2027 (expected)*
  PhD Candidate, Computer Science and Engineering East Lansing, USA
  - Advisor: Dr. Hui Liu, Dr. Jiliang Tang
- **Fudan University** *2019 – 2021*
  Master of Applied Statistics, School of Data Science Shanghai, China
- **Fudan University** *2015 – 2019*
  Bachelor of Sociology, Department of Sociology Shanghai, China

Experiences

- **TikTok Inc.** *Aug 2026 – Present*
  Research Scientist Intern San Jose, CA
  - Developed self-evolving agents for generative recommendation.
  - Built an integrated framework for modeling user preferences and feedback signals with long-tail distributions.
  - Conducted research on continual learning for large language models.
- **ByteDance Inc.** *May 2026 – Aug 2026*
  Research Scientist Intern, Ark_AML San Jose, CA
  - Developed an LLM-agent harness framework for evolutionary skill optimization, supporting relevant-skill retrieval, skill performance attribution, and performance signal fusion.
  - Designed a graph-based dependency modeling method to capture interactions among evaluation dimensions, improved the consistency and interpretability of automated evaluation.
  - Built automated evaluation generation pipeline for optimizing agent skills, enlarge the supervision signals to help RL.
- **Amazon Inc.** *Aug 2025 – Nov 2025*
  Applied Scientist Intern Seattle, USA
  - Research Topics: Foundational AI for Stores; Conversational Generation and Evaluation
- **TAL Education Group** *Jun 2024 – Aug 2024*
  Summer Intern Beijing, China

◦ Research Topics: Applications of Large Language Models; Intelligent Tutoring Systems for Elementary Math

- **Data Science and Engineering Lab, Michigan State University** *Aug 2022 – Present*
  Research Assistant, Department of Computer Science and Engineering East Lansing, USA
  ◦ Research Areas: Large Language Models, AI for Education, Computational Social Science
  ◦ Advisors: Dr. Jiliang Tang and Dr. Hui Liu

- **Department of Computing, The Hong Kong Polytechnic University** *May 2023 – Aug 2023*
  Research Intern Hong Kong
  ◦ Research Topics: Large Language Models; Graph Data Mining

Publications

* indicates equal contribution.

- Designing an AI-Assisted Feedback System Customizable to Faculty Style: A Design-Based Research Approach
  Danielle Silvaggio, **Kaiqi Yang**, Nickolina Yankova, Yasemin Copur-Gencturk, Jiliang Tang
  AI in Mathematics Education: Research and Practice. Springer (in press; expected 2027).

- Exploring Solution Divergence and Its Effect on Large Language Model Problem Solving
  Hang Li, **Kaiqi Yang**, Yucheng Chu, Hui Liu, Jiliang Tang
  Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026.

- Exploring Prompt Injection Attacks on LLM-Based Automatic Grading Systems
  Hang Li*, Fedor Filippov*, Yuling Lin, Pengfei He, **Kaiqi Yang**, Yucheng Chu, Yingqian Cui, Hui Liu, Jiliang Tang
  Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026.

- Designing Professional Development in Mathematics Teacher Education with AI as a Facilitator: A Roadmap Grounded in High-Quality Professional Development Features
  Nickolina Yankova, Jinhyo Cho, Hang Li, **Kaiqi Yang**, Yasemin Copur-Gencturk, Jiliang Tang
  Journal of Mathematics Teacher Education, 2026.

- A LLM-Driven Multi-Agent Systems for Professional Development of Mathematics Teachers
  Kaiqi Yang*, Hang Li*, Yucheng Chu, Ahreum Han, Yasemin Copur-Gencturk, Hui Liu
  International Conference on Artificial Intelligence in Education (AIED), 2026.

- Optimizing In-Context Demonstrations for LLM-based Automated Grading
  Yucheng Chu, Hang Li, **Kaiqi Yang**, Yasemin Copur-Gencturk, Kevin Haudek, Joseph Krajcik, Jiliang Tang
  International Conference on Artificial Intelligence in Education (AIED), 2026.

- From Flat to Structural: Enhancing Automated Short Answer Grading with GraphRAG
  Yucheng Chu, Haoyu Han, Shen Dong, Hang Li, **Kaiqi Yang**, Yasemin Copur-Gencturk, Joseph Krajcik, Namsoo Shin, Hui Liu
  International Conference on Educational Data Mining (EDM), 2026.

- Confusion-Aware Rubric Optimization for LLM-based Automated Grading
  Yucheng Chu, Hang Li, **Kaiqi Yang**, Yasemin Copur-Gencturk, Joseph Krajcik, Namsoo Shin, Jiliang Tang
  International Conference on Educational Data Mining (EDM), 2026.

- Think-Before-Speak: From Internal Evaluation to Public Expression in Multi-Agent Social Simulation
  Kaiqi Yang, Tai-Quan Peng, Sanguk Lee, Hui Liu
  Presented at KDD'26 Workshop on "Agentic AI for Scientific and Societal Advances", 2026.

- Beyond Partisan Leaning: A Comparative Analysis of Political Bias in Large Language Models
  Tai-Quan Peng, **Kaiqi Yang**, Sanguk Lee, Hang Li, Yucheng Chu, Yuping Lin, Hui Liu
  Journal of Information Technology & Politics, 2026.

- LLM-based Automated Grading with Human-in-the-Loop
  Hang Li*, Yucheng Chu*, **Kaiqi Yang**, Yasemin Copur-Gencturk, Jiliang Tang
  IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE), 2025.

- Iterative LLM-Based Generation and Refinement of Distracting Conditions in Math Word Problems
  Kaiqi Yang, Hang Li, Yucheng Chu, Zitao Liu, Mi Tian, Hui Liu
  IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE), 2025.

- Ask-Before-Detection: Identifying and Mitigating Conformity Bias in LLM-Powered Error Detector for Math Word Problem Solutions
  Hang Li, Tianlong Xu, **Kaiqi Yang**, Yucheng Chu, Yanling Chen, Yichi Song, Qingsong Wen, Hui Liu
  Annual Meeting of the Association for Computational Linguistics (ACL), 2025.

- A LLM-Powered Automatic Grading Framework with Human-Level Guidelines Optimization
  Yucheng Chu, Hang Li, **Kaiqi Yang**, Harry Shomer, Hui Liu, Yasemin Copur-Gencturk, Jiliang Tang
  International Conference on Educational Data Mining (EDM), 2025.

- Are Large Language Models (LLMs) Good Social Predictors?
  Kaiqi Yang, Hang Li, Hongzhi Wen, Tai-Quan Peng, Jiliang Tang, Hui Liu
  Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024.

- Content Knowledge Identification with Multi-Agent Large Language Models (LLMs)
  Kaiqi Yang, Yucheng Chu, Taylor Darwin, Ahreum Han, Hang Li, Hongzhi Wen, Yasemin Copur-Gencturk, Jiliang Tang, Hui Liu
  International Conference on Artificial Intelligence in Education (AIED), 2024.

- Spectral-Aware Augmentation for Enhanced Graph Representation Learning
  Kaiqi Yang, Haoyu Han, Wei Jin, Hui Liu
  ACM International Conference on Information and Knowledge Management (CIKM), 2024.

Preprints

* indicates equal contribution.

- Words to World: Benchmarking LLMs in Interactive Physical Environments
  Kaiqi Yang*, Xinnan Dai*, Yaxin Li*, Cheng Luo, Wanli Li, Mingqing Xiao, Xinyang Jiang, Daoan Zhang, Kaitao Song, Fengze Liu, Zefan Cai, Caihua Shan
  arXiv preprint arXiv:2610.15228, 2026.

- How Uncertain Is the Grade? A Benchmark of Uncertainty Metrics for LLM-Based Automatic Assessment
  Kaiqi Yang*, Hang Li*, Xianxuan Long, Fedor Filippov, Yucheng Chu, Yasemin Copur-Gencturk, Peng He, Cory Miller, Namsoo Shin, Joseph Krajcik, Hui Liu, Jiliang Tang
  arXiv preprint arXiv:2602.16039, 2026.

- Enhancing LLM-Based Short Answer Grading with Retrieval-Augmented Generation
  Yucheng Chu, Peng He, Hang Li, Haoyu Han, **Kaiqi Yang**, Yu Xue, Tingting Li, Joseph Krajcik, Jiliang Tang
  arXiv preprint arXiv:2504.05276, 2025.
- Exploring Social Desirability Response Bias in Large Language Models: Evidence from GPT-4 Simulations
  Kaiqi Yang*, Sanguk Lee*, Tai-Quan Peng, Ruth Heo, Hui Liu
  arXiv preprint arXiv:2410.15442, 2024.

Presentations

- Bringing Generative Artificial Intelligence (GenAI) to Education
  Hang Li, **Kaiqi Yang**, Yucheng Chu, Jiliang Tang
  Tutorial, International Conference on Educational Data Mining (EDM), Palermo, Italy, 2025.
- Innovative Artificial Intelligence Solutions for Improving Online Professional Development for Teachers: Multi-Agent Large Language Models
  Kaiqi Yang, Yasemin Copur-Gencturk, Ahreum Han, Hang Li, Jingxian Li, Jiliang Tang
  Symposium: "The Use of AI in Education and Research in Meaningful and Responsible Ways," Annual Meeting of the American Educational Research Association (AERA), Denver, USA, 2025.

Honors and Awards

- Student Travel Award, ACM CIKM 2024
- Outstanding Graduate Student Scholarship, School of Data Science, Fudan University 2020/21
- Outstanding Undergraduate Student Scholarship, Fudan University 2016/17/18/19

Teaching

- Teaching Assistant, CSE 482: Big Data Analysis Spring 2025, Spring 2024
  Responsibilities included grading coursework, holding office hours, and mentoring students on honors projects.
- Teaching Assistant, CSE 440: Introduction to Artificial Intelligence Fall 2024, Fall 2023
  Responsibilities included managing online discussions, holding office hours, and grading coursework.

Mentoring

- Yucheng Chu, PhD Student Dec 2023 – Jan 2025
  Project: Large Language Models for STEM Education
  Co-authored publication: AIED 2024

Service

- Conference Reviewer 2022 – Present
  Reviewed for WWW, WSDM, NeurIPS, ICLR, ICDM, ICML, ACL
- Journal Reviewer 2022 – Present
  Reviewed for TKDE, TNNLS, TKDD